fNIRS_3tasks Manual

General note on computers and network

In this section, we identify the computers used throughout the experiment and the software installed on them.

The two MacBooks (i.e., the participant computers named as Tom and Jerry) are connected to the Inria-interne network via Ethernet. Connecting via Ethernet is the most secure connection. Backup: in case Ethernet is not available, we connect the computers to Inria-exp (this is the network Jean-Marc has created for us). For instance, Ethernet is currently not available in rooms D004 and D005.

fNIRS devices (white boxes that we attach optodes to) are connected to the network Collaboration-exp which has been previously configured on the TP-Link mesh router.

We also connect Alienware to Collaboration-exp, the network configured on TP-Link mesh. This connection serves as a way to connect the fNIRS devices to the software. Additionally, we connect Alienware to the internet through Ethernet cable. This internet connection allows users to start a Webex call on Alienware and share the PsychoPy through screen share. PsychoPy has to be on the same computer as the fNIRS software because we send the triggers this way.

Below are brief descriptions of the software running on each computer:

2 MAC computers (Tom: MAC-02008474, Jerry: MAC-02008255)

1) **OBS:**

Allows recording of the selected layout, which includes the video from the external camera and the participant's screen.

2) **Webex:**

This is how the participant connects to the video conference. So that the 2 participants can see each other, and see the task they are collaborating on.

Alienware: experimenter PC

1) **Hyperscan and Aurora**

Experimenter connects fNIRS devices to Hyperscan. Aurora which automatically opens when the recording is started in Hyperscan. Both software can be found on the desktop.

2) **PsychoPy**

Experimenter presents the task that the participants will collaborate on using PsychoPy. Experimenter shares their PsychoPy screen during the Webex videoconference.

3) **Webex**

Experimenter starts the Webex video conference, and also records their screen directly of the call on Webex.

We have **NIRSite** software installed on our Alienware system, which is used to configure the cap montage—specifying the exact locations for sources and detectors. While we don't use it every time since a pre-configured montage has already been sent to Aurora, the current setup includes 16 sources and 15 detectors, with the 16th detector replaced by a short-channel bundle. Should we need to modify or create a new configuration, NIRSite allows us to load and adjust the existing montage or build one from scratch.

One Week Prior to the Experiment

a) Review questionnaire responses: Analyse the participants' responses to identify pairs who are fluent in the same language, do not know each other, and have matching availability during the week.

b) Contact participants: Use the email addresses provided in the questionnaire to send the study information and consent forms (available in the "3-tasks" drive folder). Await their confirmation of participation.

c) Visit participants: Once confirmed, schedule a visit to their office to measure their head sizes.

d) Send out reminders to the participant.

e) Prepare the equipment: Select the appropriate cap size and fit the optodes in preparation for the study.

Experiment Day (start up to 2 hours prior to the participant's arrival)

1. Hardware preparation

1.1 Transporting the hardware

Please transport the following equipment (stored in the cabinet in B030) to the testing space, which Sophie will reserve. Ensure that none of the computers are left unattended; always have another person supervising them. If needed, please remain in the testing space until the other team members arrive.

Screens, Computers, Mesh router, fNIRS

- [ ] **3 Dell screens**
  - [ ] Tom: 02008094 (Inria tags at the back)
  - [ ] Jerry: 02008841
  - [ ] Experimenter: 02008092
- [ ] **2 MAC computers**
  - [ ] Tom: (MAC-02008474)
  - [ ] Jerry: (MAC-02008255)
- [ ] **Alienware computer**
- [ ] **Mesh router**
- [ ] **Black fNIRS suitcases**

Jerry Box

Cameras, microphones

- [ ] **Logitech camera**
- [ ] **Microphone**

Camera and microphone accessories & cables

- [ ] **Cork holder** (yellow rectangular corks with space to secure the microphones in place)
- [ ] **Cable adaptor**
- [ ] **USB-C to USB-C cable** (connects screen to adaptor)
- [ ] **USB-C to USB cable** (connects camera to adaptor)
- [ ] **Batteries** (for microphone)

White fNIRS Cap preparation box

- [ ] **Measurement tape** (to check the distance in between participant and screen, participant and microphone)
- [ ] **Earbuds and signal optimisation pen** (both are used for signal optimisation)
- [ ] **Spring tops**

Others

- [ ] **Gift, food**
- [ ] **Finn the fox** (book)

1.2 Hardware checklist

- [ ] Make sure to have a chair with no wheels (so that the participant can't move during the experiment).
- [ ] Connect the screen to the adaptor using a USB-C to USB-C cable.
- [ ] Connect Logitech camera to the same adaptor.
- [ ] Connect the adaptor to the Mac. (adaptor connects to the Mac via USB-C port)
- [ ] Insert the newly charged batteries inside the device.
- [ ] Place the screen and microphones according to the standard distance. (actual distance to be defined)
- [ ] Put the microphone on the cork holder.
- [ ] The device has two microphones facing opposite directions. Make sure both microphones are at 90 instead of 120 (this is set at the tip of the microphone) (insert a picture to make it more clear).
- [ ] Position the device so that one of the microphones faces the mouth of the speaker. Put an X made of tape on the table where the microphone is so that we know that it hasn't been moved.
- [ ] Turning on the microphone: Locate the power button on the left side of the device and slide it towards the power icon. A 'Z' symbol will appear on the screen.

- ☐ Selecting where to save your data: Press the folder button (button 1). The default storage location, FOLDER01, will be selected. Press the same button again to proceed to the recording page.

- ☐ Starting the recording: Tap the red circle button (record button) to initialise the recording. The file name "STE-000.wav" will be displayed on the screen. Then click on the rec button a **second time** to start the recording.

- ☐ Verify Recording: Confirm that the time count appears on the screen and is changing, indicating that the recording is in progress.

- ☐ Ending the recording: This is a bit counterintuitive. Clicking on the rec button again won't stop the recording. To stop the recording, **press the square button**, which serves as the conventional stop recording button.

1) Troubleshooting:

What if something went wrong and you want to record again? No worries. If you encounter issues and need to start a new recording, press the folder button twice and then tap the record button. The file name will increment to "STE-001.wav," indicating the new recording. Remember to press the record button again to start and stop the recording with the square button.

2) Taking out the SD card:

Before removing the SD card, power off the device by following the same procedure as turning it on. Once powered off, press the SD card on the right side of the device to release it from the holder. Place the SD card in the adapter for further use.

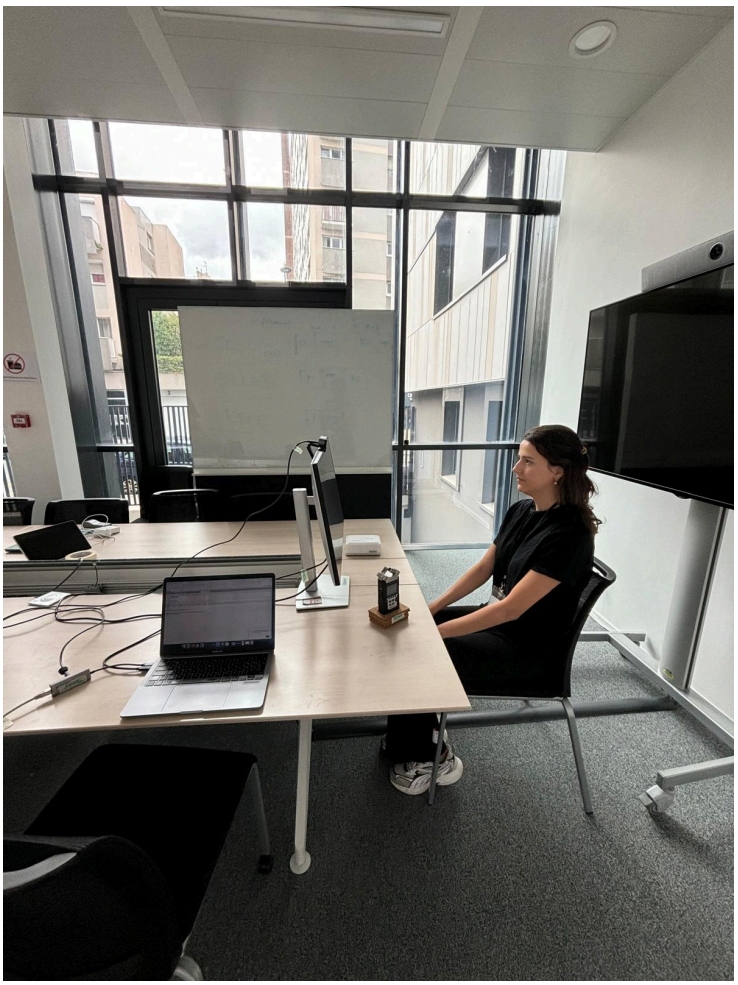
3) Copying data on participant computer

Verify that the files exist in the STEREO folder under FOLDER01. Ensure to copy the entire STEREO folder to prevent data loss due to potential file errors.

4) Important step

After copying the data, put on a new SD card (although we copied the data, it is important to keep them on the used SD card until we are sure there's no problem). Additionally, remove the batteries and charge them. Ensure that the stamina button, located above the battery holder, is switched off (indicated by the arrow) to prevent accidental activation.

5) Microphones are ready for the next experiment!



2. Software preparation

2.1 Run PsychoPy

These steps pertain only to the Experimenter room. First, run the Psycophy script. Then, open the Aurora software and verify that the triggers are visible. Once you confirm this, you're ready to start the experiment at your convenience. The initial screen is set up to allow you to skip it and begin the experiment (please wait for further instructions).

2.2 Start the Webex Call

A) Instructions for Experimenter Room

- ☐ Login to your Webex account and start a call
- ☐ Mute yourself

- ☐ Turn off the camera
- ☐ Start screen sharing and select PsychoPy screen

2.3 Join the Webex Call

B) Instructions for Tom's Room

- ☐ **Turn off Self-View**
- ☐ Click on **Stop Video** to hide Jerry's video feed from the screen.
- ☐ Scroll up to enlarge the experimenter's shared screen. This will push Jerry's video and gray screen out of view, so that only the experimenter's screen share is visible to Tom.

C) Instructions for Jerry's Room

- ☐ **Turn off Self-View**
- ☐ Click on **Stop Video** to hide Tom's video feed from the screen.
- ☐ Scroll up to enlarge the experimenter's shared screen. This will push Jerry's video and gray screen out of view, so that only the experimenter's screen share is visible to Tom.

2.3 Setup OBS in Tom and Jerry's rooms

You may find the OBS icon on the Dock menu and click on it to launch OBS. We have three inputs to set on OBS: audio capture from camera, video capture from camera, and screen capture from the Dell screen. Here is how you set the 3 inputs:

a) You see a **Sources** panel, click on the plus sign on the panel, then select **Audio Input Capture**. A window will pop up where the create new option is automatically selected. Right underneath, change the automatic name to Webex screen. Once you click ok for the creation of the new source, a Properties window will pop up where you need to select the device as **Logitech BRIO** and click ok.

b) On the Sources panel, click on the plus sign again. This time select **Video Capture Device** and click ok. A window will pop up where the create new option is automatically selected. Right underneath, change the automatic name to Webex screen. Then you click ok for the creation of the new source, a Properties window will pop up where you need to set the device as **Logitech BRIO** and click ok. Set the webex screen so that it is half of the screen.

c) On the Sources panel, click on the plus sign a third time. Once you click ok for the creation of the new source, a Properties window will pop up where you need to select **macOS Screen Capture** and click ok. Select **DELL S2721** as **Display.** Set the Webex screen so that it is half of the screen.

Giving final instructions to the participants
Pressing run recording on devices order

- Microphone

- OBS
- Webex starts
- Psychopy script is run
- Check triggers on Aurora and start Aurora recording
- Psycopy verification screen is skipped, experiment continues
- Track the signal on Aurora software from Alienware screen (dell screen), watch the participants on webex screen

3. When Participant Arrives

1) There are two types of optodes: sources and detectors. Sources are indicated with red and detectors with blue. Plug in sources and detectors to the NIRX device and turn on NIRX devices

2) Open Hyperscan software found on Alienware desktop
    a) Click on the "+"sign to add a device. Then pick the device 2233_0481_A. Refresh if no device shows.
    b) Click on the "+"sign to add a device. Then pick the device 2230_0462_A. Refresh if no device shows.
    c) The software will then ask you to pick configuration. Pick the configuration GuessBrain_GOOD
    d) Click on the plug button to connect each of the devices. When the status is ready, put on the caps on the subject:

Capping Instructions
Participant script

    i) Position the cap on the participant: Place the cap on the participant's head, starting from the front and pulling it gently towards the back.
    ii) Check alignment of the cap: Measure the distance from each ear to the center of the head (midline) and ensure both sides are equal. Ensure that the distance between the nasion (bridge of the nose) and the cap’s midline is equal on both sides.

    e) Put the cap from front to the back, and measure the distance from one ear to the midline, should eb the same from both sides. Also the distance should be same in between nose and
    f) When the cap is on, start calibrating by clicking on the run button. This will show you which channels need signal improvement. This will automatically open 2 windows from Aurora software, one for each. For signal optimisation, change the spring tops to pressure 2 or 3 (0 means no spring so no pressure).

3) Check that you are able to see the triggers on Aurora software. Click on configuration (GuessBrain) > Edit > LSL stream names. Type in trigger in stream name [confirm] > Test connection. Our stream name is "Trigger"

4) Start the PsychoPy experiment from Alienware.

4. Press record

Start recording on OBS
Start recording on

After the measurement checklist

- ☐ Charging the fNIRS device
- ☐ Charging 2 MACs
- ☐ Charging microphone batteries (see 5)Microphone Manual)
- ☐ Washing the caps

5)Microphone Manual

6) Make sure the device has fully charged batteries inserted.

7) Put the microphone on the cork holder.

8) The device has two microphones facing opposite directions. Make sure both microphones are at 90 instead of 120 (this is set at the tip of the microphone).

9) Position the device so that one of the microphones faces the mouth of the speaker. Put an X made of tape on the table where the microphone is so that we know that it hasn't been moved.

10) Turning on the microphone:
Locate the power button on the left side of the device and slide it towards the power icon. A 'Z' symbol will appear on the screen.

11) Checking the battery:
Verify the battery level by checking the battery icon located in the top right corner of the screen.

12) Selecting where to save your data:
Press the folder button (button 1). The default storage location, FOLDER01, will be selected. Press the same button again to proceed to the recording page.

13) Starting the recording:
Tap the red circle button (record button) to initialise the recording. The file name "STE-000.wav" will be displayed on the screen. Then click on the rec button a **second time** to start the recording.

14) Verifying Recording:
Confirm that the time count appears on the screen and is changing, indicating that the recording is in progress.

15) Ending the recording:

This is a bit counterintuitive. Clicking on the rec button again won't stop the recording. To stop the recording, **press the square button**, which serves as the conventional stop recording button.

16) Troubleshooting:

What if something went wrong and you want to record again? No worries. If you encounter issues and need to start a new recording, press the folder button twice and then tap the record button. The file name will increment to "STE-001.wav," indicating the new recording. Remember to press the record button again to start and stop the recording with the square button.

17) Taking out the SD card:

Before removing the SD card, power off the device by following the same procedure as turning it on. Once powered off, press the SD card on the right side of the device to release it from the holder. Place the SD card in the adapter for further use.

18) Copying data on participant computer

Verify that the files exist in the STEREO folder under FOLDER01. Ensure to copy the entire STEREO folder to prevent data loss due to potential file errors.

19) Important step

After copying the data, put on a new SD card (although we copied the data, it is important to keep them on the used SD card until we are sure there's no problem). Additionally, remove the batteries and charge them. Ensure that the stamina button, located above the battery holder, is switched off (indicated by the arrow) to prevent accidental activation.

20) Microphones are ready for the next experiment!